Essays

|!|BCEEA0070000001|!| ★★ **Cross topic**

Nitrogen is a mineral essential to the human body. Give an account on how humans obtain nitrogen from diet and the assimilation of nitrogen into body proteins. (12 marks)

##

Source of nitrogen

- Humans obtain nitrogen from proteins in food. (1)

Digestion of proteins in food

- Food containing proteins is mechanically broken into small pieces in the mouth cavity and the stomach (chewing by the teeth and churning in the stomach) (1)
- Chemical digestion of proteins starts in the stomach. Gastric juice contains proteases and hydrochloric acid. (1)
- The protease pepsin in the stomach catalyses the breakdown of proteins into peptides. (1)
- In the small intestine, pancreatic proteases catalyse the breakdown of proteins into peptides and amino acids. (1)
- Proteases on the cell membranes of the epithelial cells in the intestinal wall catalyse the breakdown of peptides into amino acids. (1)

Absorption of amino acids

- Amino acids in the small intestine are absorbed by diffusion and active transport into the blood capillaries of the villi. (1)
- The amino acids are carried by the bloodstream to different parts of the body for cells to use. (1)

Assimilation of amino acids

- Amino acids are used as raw material to make organelles (e.g. cell membrane and cytoplasm), cells, muscles, connective tissue, etc. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Applying concepts	9
Communication	3

© 2020 Aristo Educational Press Ltd.

|!|BCEEB0070000002|!| ★

John ate some fried pork chops at dinner. Discuss how the lipids in the pork chops are digested and absorbed, and how the lipids may become his body fat. (12 marks)

##

Digestion of lipids (Max. 3)

- In the duodenum, bile and pancreatic juice help the digestion of lipids. (1)
- Bile, which is secreted by the liver, contains bile salts. (1)
- Bile salts cause the emulsification of lipids into smaller droplets. (1)
- Pancreatic juice, which is secreted by the pancreas, contains pancreatic lipase. (1)
- The small oil droplets are digested by pancreatic lipase into fatty acids and glycerol. (1)
- Digestion of lipids is optimum under alkaline condition. (1)

Absorption of fatty acids and glycerol (Max. 3)

- Fatty acids and glycerol molecules pass through the epithelium of the villi by diffusion (1)
- and then recombine to form lipid molecules (1)
- which then enter the lacteal (1)
- and are carried away by the lymph. (1)

Assimilation of lipids (Max. 3)

- After entering the lacteal, lipid molecules are carried by the lymph through the lymph vessels. (1)
- The lymph vessels eventually open into two large veins near the neck region, returning lymph to the blood. (1)
- The lipid molecules are carried by the bloodstream to different parts of the body and are taken up by cells for metabolism. (1)
- Excess lipid molecules are stored as adipose tissues around internal organs or subcutaneous fat under the skin. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Applying concepts	9
Communication	3

© 2020 Aristo Educational Press Ltd.

|!|BCEEA0070000003|!| ★★

Describe how the structure of different parts of the alimentary canal, starting from the stomach, enables humans to perform digestion and absorption (details of the enzyme reactions are not required). (11 marks)

##

Alimentary canal

- The whole alimentary canal is made up of circular muscles and longitudinal muscles. (1)
- Alternate contraction and relaxation of these two layers of muscles enables peristalsis. (1)

Stomach (max. 2)

- The stomach is a muscular sac. (1)
- It can expand to accommodate variable amounts of food. (1)
- Gastric glands in the inner wall of the stomach secrete gastric juice for the digestion of proteins. (1)
- The muscular wall of the stomach contracts rhythmically (called churning). Churning helps break food into smaller pieces and mix food with gastric juice. (1)

Small intestine (max. 4)

- The membranes of epithelial cells of the small intestine contain carbohydrases and proteases for the digestion of disaccharides and peptides. (1)
- The small intestine is very long. (1)
- The inner surface of the small intestine is highly-folded / contains many villi and microvilli. (1)
- This increases the surface area for absorption. (1)
- The epithelium of each villus is thin / one-cell thick. (1)
- This shortens the distance of absorption. (1)
- Each villus contains a network of capillaries and a lacteal. (1)
- This enables rapid absorption / transport of digested food molecules. (1)
- The inner surface of the small intestine secretes mucus (1)
- for lubrication /ease of passage of food / preventing self-digestion. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

© 2020 Aristo Educational Press Ltd.

|!|BCEEA0070000004|!| ★★ **Cross topic**

The ileum is an organ. In complex organisms, cells are organized into tissues which are organized into organs. Based on your biological knowledge, discuss the roles of the different tissue layers of the ileum in digestion and absorption. (11 marks)

##

Epithelium of the inner wall of the ileum (Max. 4)

- It is the innermost layer which is in contact with food. (1)
- It is deeply folded into numerous villi to increase the surface area for absorption. (1)
- The surface of the epithelial cells of villi bears microvilli which further increases the surface area for absorption. (1)
- The epithelium of each villus is thin / one-cell thick. This shortens the distance of absorption. (1)
- The epithelial cells of villi contain many mitochondria which provide energy for active transport of certain food molecules. (1)

Blood vessels and lymph vessels (Max. 2)

- The blood vessels branch into many networks of blood capillaries, while the lymph vessels branch into many lacteals. (1)
- Each villus has a central lacteal surrounded by a dense network of capillaries. These vessels carry the absorbed nutrients away rapidly. (1)

Circular muscles and longitudinal muscles (Max. 2)

- Alternate contraction and relaxation of these two layers of muscles enables peristalsis. (1)
- Peristalsis moves food along the ileum. (1)
- Peristalsis helps mix food and digestive juice together. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

© 2020 Aristo Educational Press Ltd.